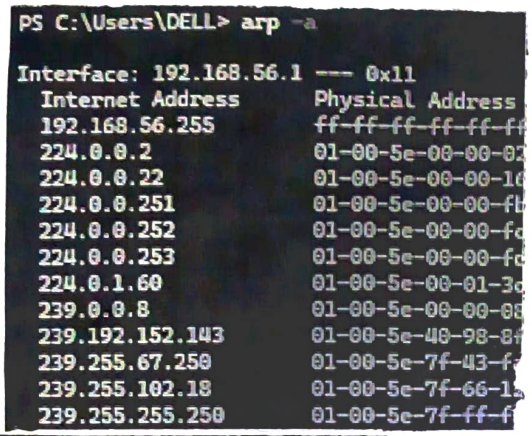
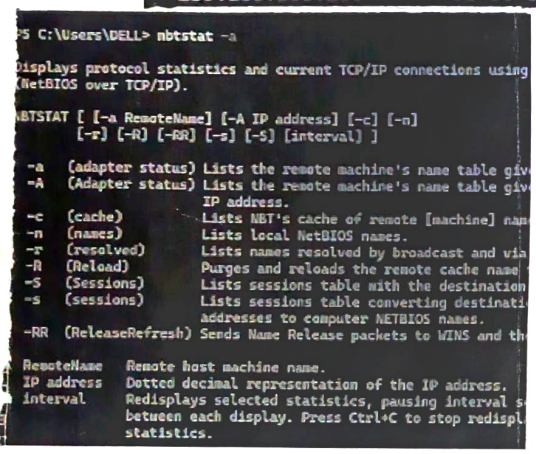


```
PS C:\Users\DELL> arp -a

Interface: 192.168.56.1 --- 0x11
  Internet Address      Physical Address
  192.168.56.255        ff-ff-ff-ff-ff-ff
  224.0.0.2             01-00-5e-00-00-02
  224.0.0.22            01-00-5e-00-00-16
  224.0.0.251           01-00-5e-00-00-fb
  224.0.0.252           01-00-5e-00-00-fc
  224.0.0.253           01-00-5e-00-00-fd
  224.0.1.60            01-00-5e-00-01-3c
  239.0.0.8             01-00-5e-00-00-08
  239.192.152.143       01-00-5e-40-98-8f
  239.255.67.250        01-00-5e-7f-43-fa
  239.255.102.18        01-00-5e-7f-66-12
  239.255.255.250       01-00-5e-7f-ff-fa
```



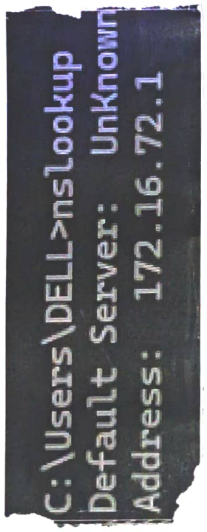
```
PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and current TCP/IP connections using
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table giv
  -A   (Adapter status) Lists the remote machine's name table giv
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] nam
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via
  -R   (Reload)         Purges and reloads the remote cache name
  -S   (Sessions)       Lists sessions table with the destination
  -s   (sessions)       Lists sessions table converting destinati
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and th

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval s
               between each display. Press Ctrl+C to stop redispl
               statistics.
```



```
C:\Users\DELL>nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

Exp. No : 1

16/07/2026

EXP. NO. 1

BASIC NETWORKING COMMAND

AIM :

To implement Basic Networking Commands in linux and windows operating system.

BASIC NETWORKING COMMANDS :

- arp -a :

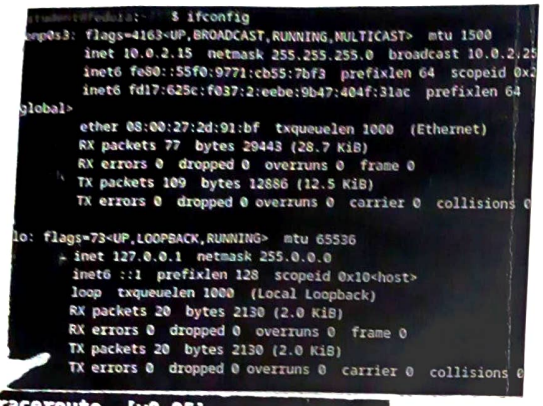
ARP is short form of address resolution protocol.

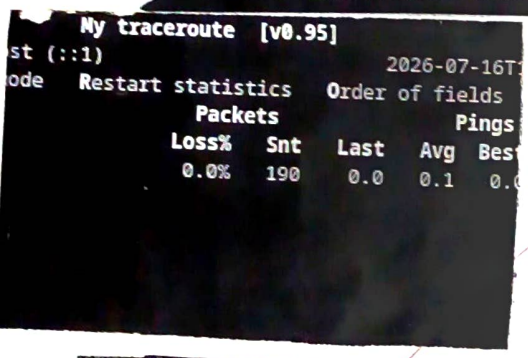
- hostname :

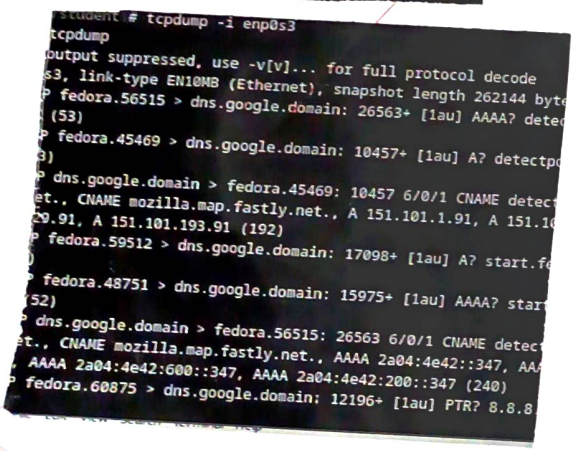
This is the simplest of all TCP/IP commands. It simply displays the commands.

- ipconfig /all :

This command displays detailed configuration, information about the connection including Router, Gateway, DNS, DHCP.







- netstat - a :

This command helps solve problem with NetBIOS name resolution for TCP/IP connection. It is a command line tool for monitoring network incoming and outgoing.

- nslookup :

name server lookup is a tool used to perform DNS lookups in display DNS details, such as IP address of a particular computer.

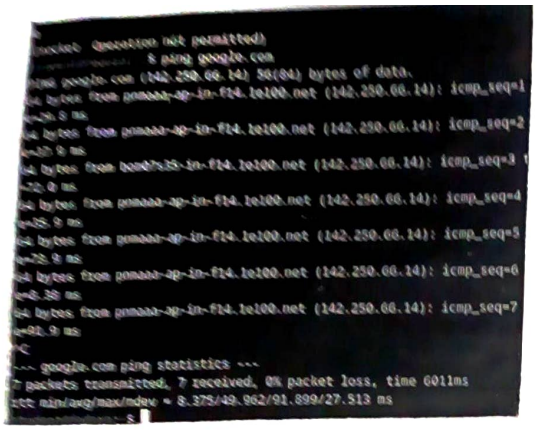
eg : nslookup www.google.com

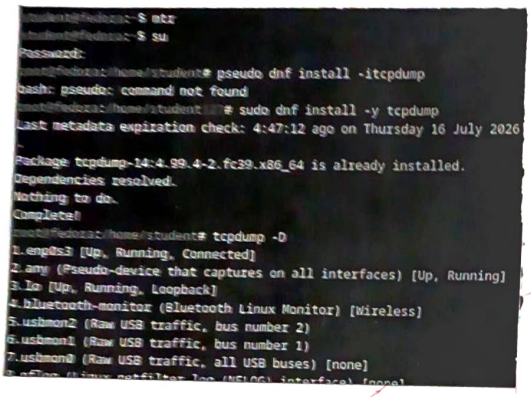
- Pathping :

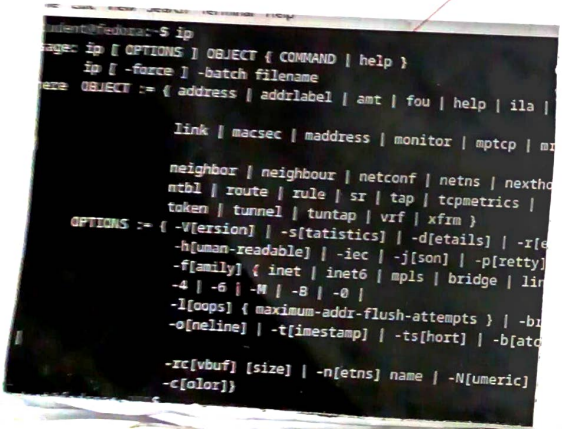
Pathping is unique to window's and is basically a combination of tracert commands.

- ping :

ping is a command to test connectivity needs.







- Route :

Route Commans is used to show / manipulate the IP routing.

LINUX NETWORKING COMMANDS :

- ip :

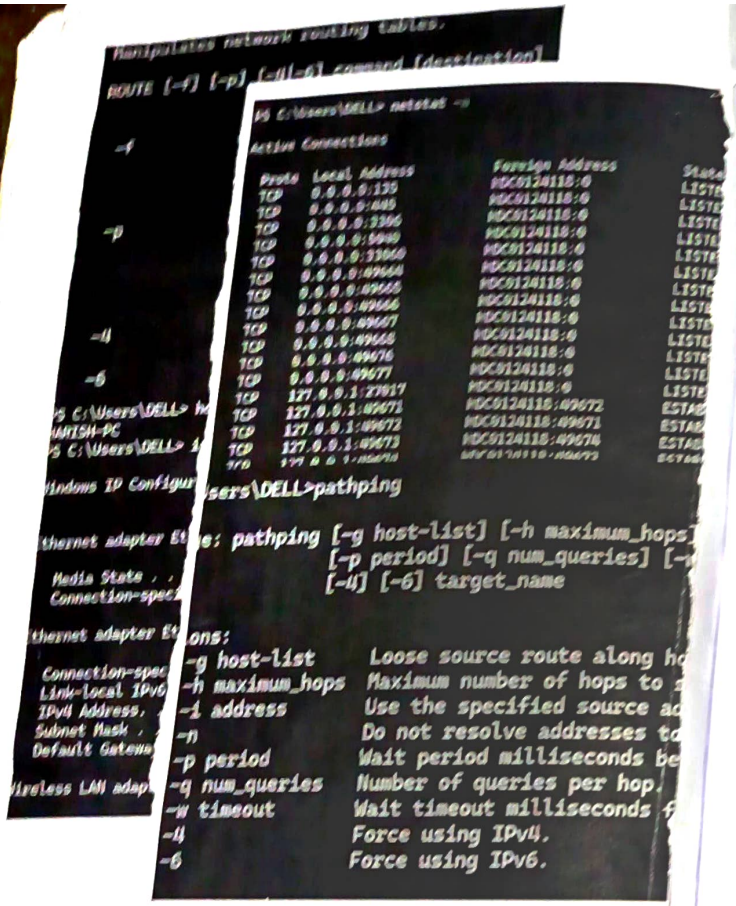
The ip command can show address information, manipulate routing, plus, display network.

ip< options ><object><command>

- To show the Ip addresses assigned to an interface on your server.
- To assign an Ip to an interface like enps03.

2) Ifconfig

The ifconfig command was a staple in many sysadmin's tool belt for configuring and trouble - shooting network.



- mtr

MTR is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroute commands.

- tcpdump

The tcpdump command is designed for capturing and displaying packets.

- ping :

Ping is a tool that verifies IP-level connectivity to another ICP/IP Computer by sending Internet control Message protocol.

Result:

Basic networking Commands in both linux and windows operating System is successfully executed & verified